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**Biofuels: Catalysts for India's
Energy Transition and Global Leadership**

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**Mission LIFE: A Visionary
Initiative for a Sustainable Future**

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**MISSION LIFE
& BIOFUELS**

Introduction

In India, the growing concerns about long-term energy security, depleting fossil fuel reserves and their environmental impact have created a greater stimulus to promoting renewable energy, particularly in sectors where larger gains are possible. India's power generation sector remains heavily reliant on coal, which continues to contribute significantly to its energy mix. In 2023, coal accounted for 59% of India's primary energy supply, despite the country's efforts to expand its renewable energy capacity. This dependence on coal is a major contributor to greenhouse gas emissions, air pollution, and other environmental hazards. The extensive use of coal exacerbates climate change and hinders progress toward achieving global sustainability targets.

Government of India launched Mission LIFE (Lifestyle for Environment) an ambitious initiative to promote environmental sustainability and address the global climate crisis (1). Mission LIFE provides a framework for integrating eco-friendly practices into daily life, while the green energy transition offers a pathway to reducing carbon emissions, improving energy security, and promoting sustainable development. By embracing renewable energy, clean coal technologies, and biofuels, India can reduce its reliance on fossil fuels and move toward a cleaner, more sustainable energy future. At its core, Mission LIFE promotes for conserving natural resources, saving water and energy, and fostering a sustainable future. The strategies seek to engage individuals, industries, and policymakers in a collective responsibility to protect the environment through sustainable practices.

India's energy transition is already underway, with a concerted effort to increase the share of renewable energy in its energy mix. India has made significant progress in expanding its renewable energy capacity, which has grown substantially in recent years. However, the rapid rise in energy demand, driven by population growth and industrialization, means that India's reliance on coal is likely to persist for the next few years. India's green energy transition is aligned with the Panchamrit principles announced at COP26, which outline a roadmap for achieving climate goals. These principles include:

- Achieving 500 GW of non-fossil energy capacity by 2030
- Reducing carbon intensity by 45% by 2030
- Achieving net-zero emissions by 2070

Biofuels, derived from agriculture residues and organic waste, offer a promising solution to India's energy challenges. Biofuels such as bio-pellets, bioethanol, biodiesel and bio-CNG can help reduce the country's dependence on fossil fuels and lower greenhouse gas emissions. These renewable energy sources can be produced locally, supporting rural development and providing a sustainable way to manage agricultural and organic waste. By integrating biofuels into the country's energy mix, India can reduce its reliance on coal and petroleum, contributing to both climate change mitigation and energy security. Additionally, biofuels can help manage agricultural waste, creating a circular economy that benefits both the environment and the economy.



Biofuels potential from Agricultural Feedstocks

Biomass is one such important source of renewable energy in India which can generate considerable energy savings in the sector. As per the MNRE study report, about 774 million tons (MT) of crop residue is generated per year in India. The total yearly surplus crop residues are reported to be approximately 228 million tons (MT). Additionally, agricultural waste is produced from a broad range of subsectors, including agro-processing industries (paper and pulp production, rice mills, oil mills, sugarcane processing, distilleries, and other food and food processing industries). The State-wise annual production and its surplus quantity are given in Table 1. The surplus, if used effectively in a cleaner energy form, can contribute significantly to meeting the overall energy needs of the country in a sustainable manner.

Table 1: State-wise Total Cropped Area Production, Biomass Generation and Surplus Biomass Production in India

States/UTS	Total crop Area (Million Hectares)	Total crop Production (Million Tonnes)	Total Biomass Generation (Million Tonnes)	Surplus Biomass Potential (Million Tonnes)
Andhra Pradesh	7.36	28.62	40.01	17.09
Arunachal Pradesh	0.33	0.58	0.75	0.17
Assam	3.4	8.93	12.57	2.54
Bihar	7.28	32.5	32.57	7.98
Chhattisgarh	5.47	8.71	12.99	2.65
Goa	0.14	0.24	0.45	0.23
Gujarat	9.67	32.27	50.24	21.74
Haryana	6.6	27.17	36.24	10.91
Himachal Pradesh	0.77	1.51	2.74	0.57
Jammu & Kashmir	0.96	1.77	3.24	0.65
Jharkhand	1.96	3.32	5.31	1.2
Karnataka	10.94	51.34	34.09	14.05
Kerala	1.3	4.79	8.58	6.04
Madhya Pradesh	23.7	43.81	70.23	19.93
Maharashtra	21.07	86.48	52.54	21.49
Manipur	0.34	1.12	1.14	0.48
Meghalaya	0.25	0.91	1.37	0.56
Mizoram	0.05	0.13	0.13	0.02
Nagaland	0.44	1.37	1.37	0.44
Odisha	4.45	7.88	11.84	2.23
Punjab	7.17	37.88	53	22.25
Rajasthan	31.93	32.11	59.5	10.21
Sikkim	0.08	0.12	0.23	0.04
Tamil Nadu	8.96	47.92	52.14	12.22
Telangana	9.38	18.57	33.62	13.76
Tripura	0.35	0.97	1.41	0.25



Uttar Pradesh	24.19	246.66	124.69	21.6
Uttarakhand	1	8.05	3.55	0.72
West Bengal	8.49	38.2	47.51	16.28
Andaman & Nicobar	0.04	0.08	0.2	0.13
Chandigarh	0	0	0	0
Dadra & Nagar Haveli and Daman & Diu	0.02	0.1	0.07	0.02
Puducherry	0.02	0.27	0.12	0.04
Total	198.11	774.38	753.3	228.49

MNRE Report, Submitted by: Centre for Energy Studies, Administrative Staff College of India (ASCI), Hyderabad (2021).

Uses of biofuels

The biofuels in all three physical forms namely; solid, liquid and gaseous forms are being used for different applications across sectors. A few important applications are as below.

Cooking

Densified biofuels bio-pellets or briquettes is considered a clean and sustainable cooking fuel using gasifier-based cook stoves. In India, wood-fired stoves and dried cow dung burning are being replaced with pellet stoves, improving sanitation and indoor air quality in rural India. Both domestic and commercial cookstoves are successfully using bio-pellets and briquettes in India.

Heating

Biomass pellets/briquettes can be used directly for heat generation and thermal applications in industries, these include.

- Furnaces, dryers and kilns that can use bio-pellets as a replacement for conventional fuels in several manufacturing sectors like food processing, chemical, pharma, metal processing and agro-processing industries etc.
- Boilers can use bio pellets as a fuel to produce steam or hot water (e.g. the steam produced by boilers can be used for space heating, process heating, or electricity generation through co-firing)

Transportation

Biofuels in liquid and gaseous forms namely, Compressed Biogas (CBG), bioethanol and biodiesel are used as a transportation fuel. Thrust is given to advanced 2G biofuels and Compressed Bio-Gas plants (CBG) technologies and including conversion of agricultural residues/waste which can be converted to ethanol and bio-CNG.

Electricity Generation

Bio-pellets can be used as fuel in internal combustion engines or steam turbines to generate electricity. The following examples are systems for electricity generation from bio-pellets

- Biomass Combustion & Co-firing

This is the most used technology for converting biomass into thermal and electrical energy



forms. The cycle used is the conventional Rankine cycle with biomass being burnt in high-pressure boiler to generate steam and operating a steam turbine with generated steam. Biomass pellets is also used along with coal in existing TPPs in cofiring mode.

- **Biomass Gasification**

Biomass gasification involves partial combustion of biomass under controlled air supply, leading to generation of producer gas. It is a thermo-chemical process in which solid biomass material is converted into a gaseous fuel by a series of processes. The producer gas can be used as fuel in the burner for producing heat energy or in IC engine for mechanical power and electricity generation.

Market Opportunities for densified biomass fuels in India

There are primarily two distinct market segments for solid biofuels in India namely.

- Micro Small and Medium Enterprises (MSMEs)
- Thermal Power Plants (TPPs)

Micro Small and Medium Enterprises (MSMEs)

Referred to as the engines of growth, the MSME sector in India plays a critical role in industrial output, exports and employment generation. Around 63.38 million MSMEs, spread over 180 clusters, provide employment to approximately 80.5 million people. Owing to the sector's enormous contribution towards the growth of the economy, the ready availability of and access to clean forms of energy at affordable prices is particularly crucial. There are many energy-intensive industries among MSMEs that still use commercial fuels like coal, oil and gas or biomass inefficiently. In most of these industries, energy costs account for nearly 30-50 percent of the total production cost. Further, these MSMEs use a wide range of locally available cheaper fuels, which together with inefficient processes and technologies lead to adverse environmental impacts, especially at the local level. For this reason, energy efficiency/fuel switching is at the centre stage for improving the competitiveness of the sector and reducing carbon emissions. Many MSMEs are exploring ways to switch to cleaner energy options, which adhere to the stringent environmental/pollution regulations. Biomass-based clean energy systems with use of pellet burners and gasification technology that has the potential to transform energy use in MSMEs in India. For thermal applications like water heating, hot air generation, drying, cooking and steam generation, comparative input energy cost Rs per 100 M-Cal heat output from different fuel types is given in Figure 1. It can be seen from Figure 1 energy cost per unit of useful energy through biomass pellet burner route is 35% cheaper than furnace oil and 45-50 % cheaper than electricity/diesel option at the current prices of diesel, electricity and biomass pellets. Thus, biomass pellets thermal option is economically viable at present and would most likely remain so for years to come. A partial list of industries, which are successfully using pellet burner/gasifier technology, is given in Table 2. Although biomass-based energy solutions have demonstrated huge potential in addressing MSME sectors' growing energy concerns, going forward, they have the potential to integrate and mainstream renewable energy at a much larger and wider scale. The innovative approach to the dissemination of pellet burners/ biomass gasifiers for thermal applications in MSME clusters involves local delivery mechanism, financing, biomass supply arrangement, operational and service support in the target sector. The



highlights of the approach include the creation of local delivery system through an entrepreneur, financing of equipment and providing processed biomass linkages along with maintenance backup facilities.

Table 2: Partial list of various MSMEs industries using pellet burners/ gasifier based on thermal energy

S. No.	Name of Industry
1.	Powder Coating Bakery
2.	Textile Dyeing
3.	Printing & Packaging
4.	Roto-Moulding
5.	Garment Industry
6.	Pharma Industry
7.	Food Processing Industries
8.	High Temperature furnaces annealing/ Hardening
9.	Dal/ Pulse processing
10.	Plastic moulding
11.	Bakeries

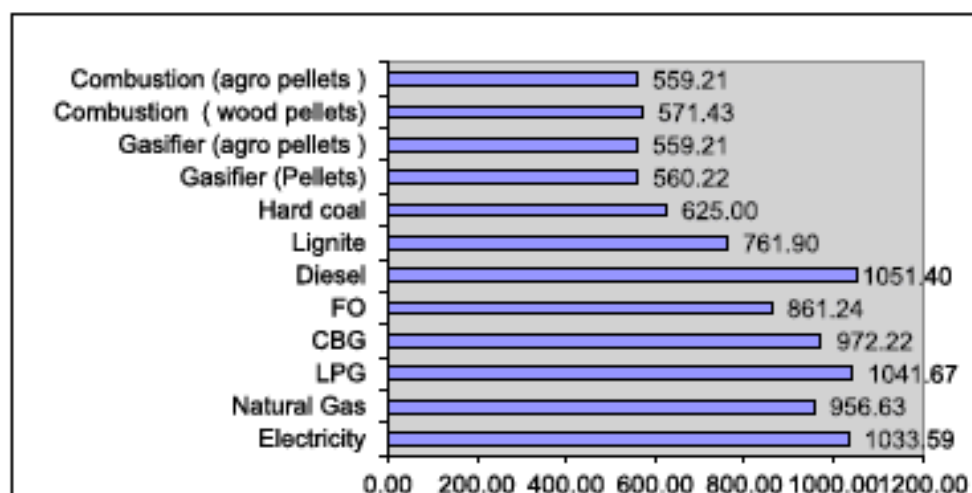


Figure 6: Comparative fuel cost Rs per 100 M-Cal useful from different fuel types

Thermal Power Plants (TPPs):

The Ministry of Power, Government of India (GoI) has initiated National Mission on use of biomass in coal based thermal power stations (SAMARTH) through co-firing in pulverized coal-fired boilers and mandated all the coal-based power plants to use 5% blend of biomass pellets along with coal. This has been initiated with an objective to create market for biomass pellets and torrefied pellets in India and to achieve the national goal to address the issue of air pollution due to farm stubble burning and to reduce carbon footprints of thermal power generation. This aims to further support the energy transition in the country and achieve its targets to move towards cleaner energy sources. The proposed National Mission on biomass will also contribute to the National Clean Air Programme (NCAP).

Approximately, 2.5 to 3.0 lakh tonnes of biomass pellets are required for 7% blending in a thermal power plant of 1000 MW capacity. The existing India's coal-based electricity production capacity is about 206 GW and annually about 972 million tonnes of coal is used in the power generation



application (2022-23). To meet 5% co-firing target, there is annual requirement of about 50 million tons of biomass pellets in TPPs across States.

Over the past few years, there has been growing trend in use of pellets by TPPs as shown in Figure 2. It can be seen about 7,81,000 MT of pellets are already co-fired in TPPs in till date of FY 2024-25 against total of 358,500 MT of pellets used in FY 23-24, in various TPPs across India. Also, cumulative capacity of Biomass co-fired is 12.58 LMT achieved till December 2024. As per SAMARTH portal data, out of total 269 TPSs in India about 65 thermal power plants have successfully implemented the co-firing policy and started firing pellets along with coal. This resulted in cumulative 15.1 lakh tons CO₂ saved through cofiring biomass pellets.

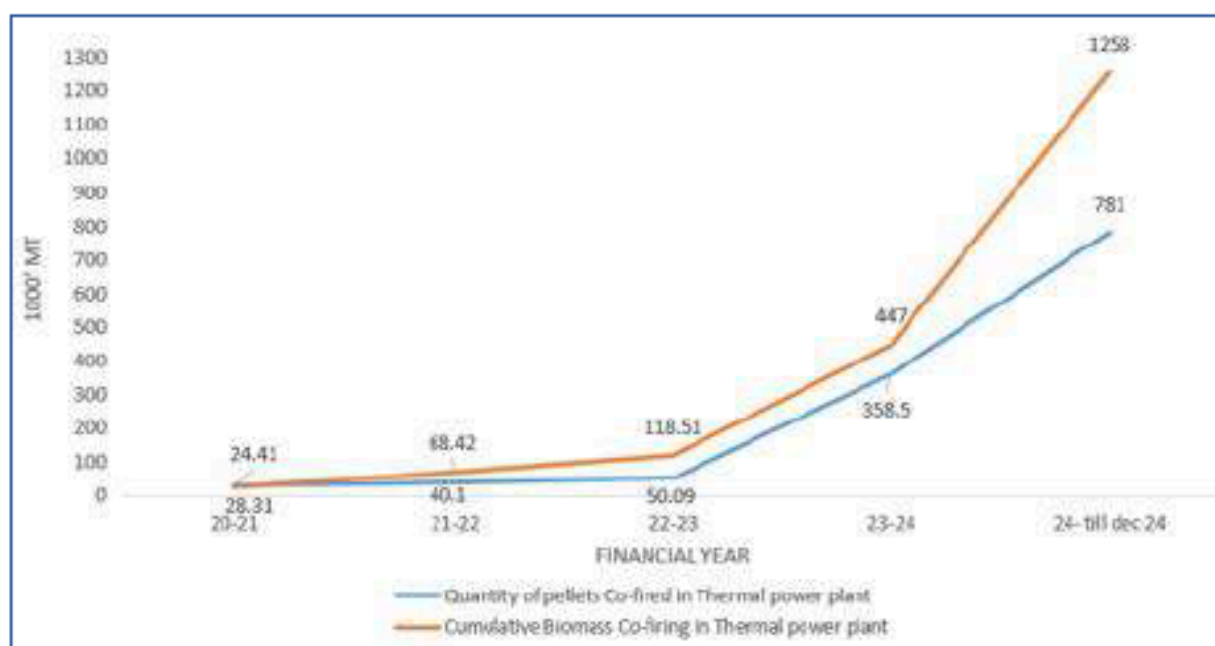


Figure 7: Year-wise quantity of Pellet Co-fired in Thermal Power Stations (TPSs)

The overall potential market size of agro-residues pellets is estimated to be about 40000 crores in thermal power plants. The Government has initiated various measures. The Ministry of New and Renewable Energy (MNRE) and Central Pollution Control Board (CPCB) have introduced Finance Assistance Schemes tailored for biomass pellet manufacturing units. The Reserve Bank of India (RBI) has accorded approval, designating 'Biomass pellet manufacturing' as an eligible activity under Priority Sector Lending (PSL).

Conclusions

The key learnings from these field experiences include

- It has been seen that many MSMEs reeling under high energy costs, are keen to shift to biomass pellets, which offer the convenience of commercial fuels like LPG, diesel, and kerosene, but at much lower costs.
- The heating applications in industries are becoming economically very attractive in applications where fossil fuels are being replaced (due to their continued price rise)
- For example, in case of thermal applications like water heating, hot air generation, drying, cooking and steam generation, cost per unit of useful energy through the gasification route is 35% cheaper than furnace oil/diesel option, and 50 % cheaper than electricity option, at the current prices of diesel, electricity and biomass fuels.



- Large market dissemination will call for supportive services e.g., servicing, maintenance, finance and capacity building at the local level etc. To cover these Energy Service Companies (ESCOs) and dealer level management should be encouraged.
- There is a need to develop sustainable fuel supply linkages, awareness and sensitizing programs and continued strengthening local delivery mechanism for large-scale adoption of these systems in various MSME clusters.

In India, biomass is already a significant source of energy and its dependence for energy needs expected to rise in future, so the contribution of biomass to the energy mix is projected to increase. As fossil-based fuel supplies dwindle, biomass will increasingly also be in demand as a substitute for fossil fuels and petrochemicals. Total demand for biomass can be expected to soar in the coming years. The issues of agri-residues include biomass collection, processing, storage and eventually its transportation to the point of ultimate utilization due to dispersed and voluminous nature of this resource, lack of robust institutional and market mechanism for efficient procurement of the required quantity of agri-residues in short span of time add additional challenge in management of this resource. Similarly, the lower energy density characteristics and life cycle emission perspective, it inevitable to use these resources at decentralized scale rather than transportation it to long distance. The lack of investment in crop residues collection and aggregation business is critical missing link in value chain creation.

The key research questions that still needs to be addressed are "How much sustainable biomass can be supplied from agriculture ecosystem considering complex interactions between its availability and human needs? 'And What is the availability of biomass, and what factors does society need to consider in managing it sustainably?' Life cycle analysis and GHG reduction potential from different processing pathways?

For addressing the challenge of crop residues management, it is crucial that district/block level specific plans be developed with emphasis on crop residues production and surplus quantity, demand of energy (fuels, electricity and drying/cooling applications) in agricultural, industries, transport and HH sectors, beside use of crop residues for soil incorporation purposes. A more innovative and integrated policy approach by combining in-situ and ex-situ approaches is needed for scalable and long-term solution of this problem. This can play an important role in achieving a net-zero carbon emissions economy by 2070. Also, to achieve the national goal to reduce GHG emissions and decarbonize power and transport sector in India.

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